Thermodynamics

Thermodynamics

Thermodynamic is a branch of physics which mainly deals with the transformation of heat into mechanical work.

Thermodynamic System: A system may be defined as a definite quantity of matter (solid, liquid or gases) bounded by some closed surface.

The simplest example of a system is a gas contained in a cylinder with a movable piston.

There are three classes of system:

Open System:

A system which can exchange matter and energy with the surroundings is called an open system.e.g. Air compressor.

Air at low pressure enters and air at high pressure leaves the system i.e., there is an exchange of matter and energy with the surroundings.

Closed System:

A system which can exchange only energy (and not matter) with the surroundings is called a closed system.e.g. Gas enclosed in a cylinder expands when heated and drives the piston outwards. The boundary of the system moves but the matter (here gas) in the system remains constant.

Isolated system:

A system which is thermally insulated and has no communication of heat or work with the surroundings is called isolated system.

Zeroth Law of Thermodynamics(Thermodynamic Equilibrium):

Statement:

The zeroth law of thermodynamics states that if two bodies A and B are each separately in thermal equilibrium with a third body C, then A and B are also in thermal equilibrium with each other.

Explanation:

We consider two systems A and B insulated from each other but in good thermal contact with a common system C. Systems A and B will attain thermal equilibrium with third system C (Fig.-01).

First Law of Thermodynamics:

Statement:

The heat added to the system is equal to the change in internal energy of the system plus the external work done by the system.

Explanation:

Let the quantity of heat added to the system be δH, the change in internal energy of the molecules be dU and the amount of external work done be δW.

Mathematically

$\delta H = dU + \delta W$ ………………….(1)

Equation (1) represents the first law of thermodynamics. All the quantities are measured in heat units.

Second Law of Thermodynamics:

The second law of thermodynamics is a general principle which places constraints upon the direction of heat transfer and attainable effiencies of heat engine.

Clausius's Statement:

It is impossible to make heat flow from a body at a lower temperature to a body at a higher temperature without doing external work on the working substance.

Kelvin's Statement:

It is impossible to get a continuous supply of work from a body by cooling it to a temperature lower than that of its surroundings.

Planck's Statement:

It is impossible to construct an engine which, working in a complete cycle, will produce no effect other than the raising of a weight and the cooling of a heat reservoir.

Kelvin-Plank Statement:

It is impossible to extract an amount of heat from a hot reservoir and use it all to do work.

Edser's Statement:

Heat flows of itself from higher to lower temperature.

Reversible Process:

A process which can be retraced in the opposite direction so that the working substance passes through exactly the same states in all respects as in the direct process is called a reversible process.

***Example**: A given mass of ice changes to water when a certain amount of heat is absorbed by it and the same mass of water changes to ice when the same quantity of heat is removed from it.*

Irreversible Process:

A process which cannot be retraced in the opposite direction is called an irreversible process. In an irreversible process the system with the surroundings can never be completely restored to its initial condition by reversing the controlling factors.

Example: *Exchange of heat between bodies at different temperatures by conduction or radiation.*

Isothermal Process:

If a system is perfectly conducting to the surrounding and the temperature remains constant throughout the process is called an isothermal process.

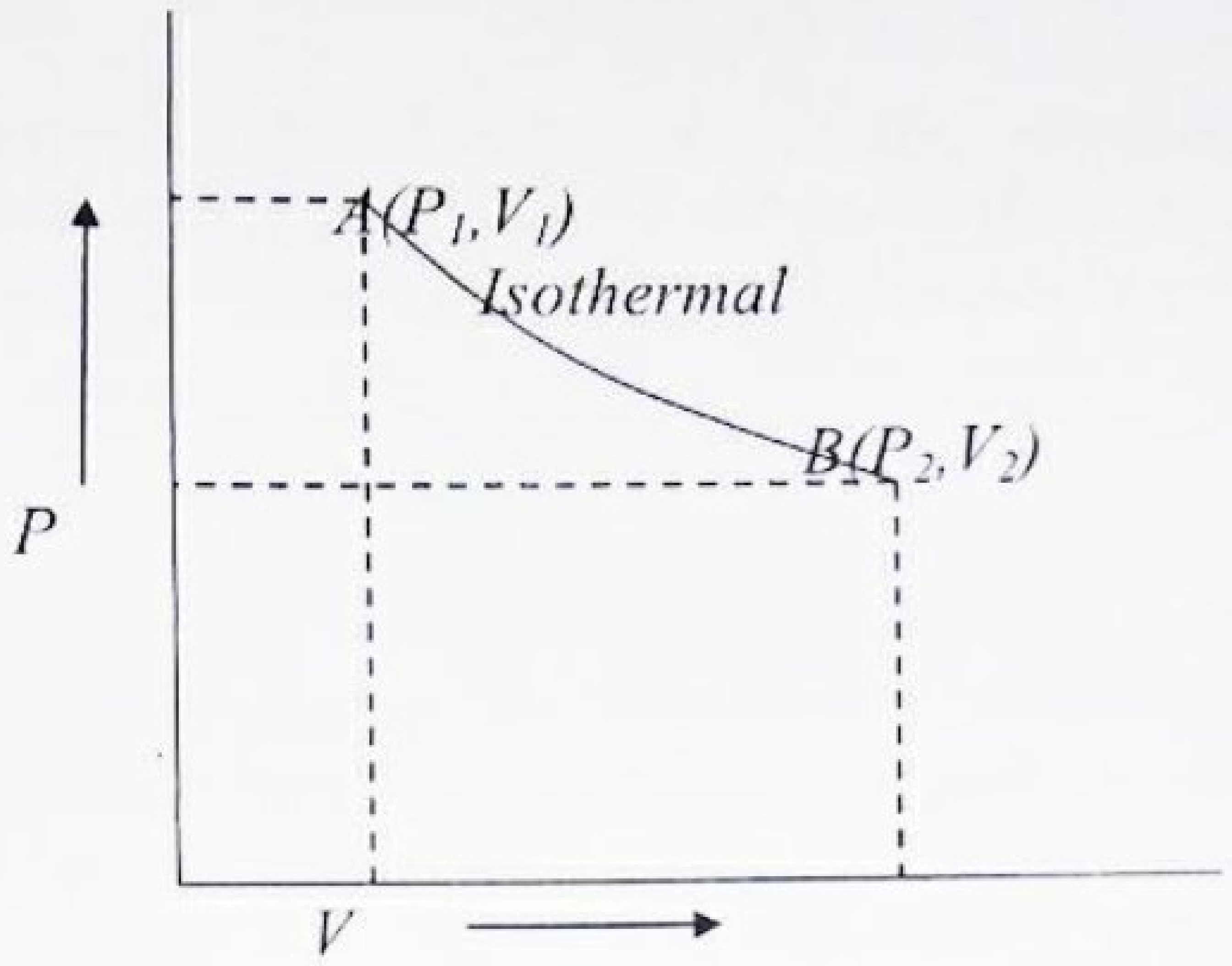


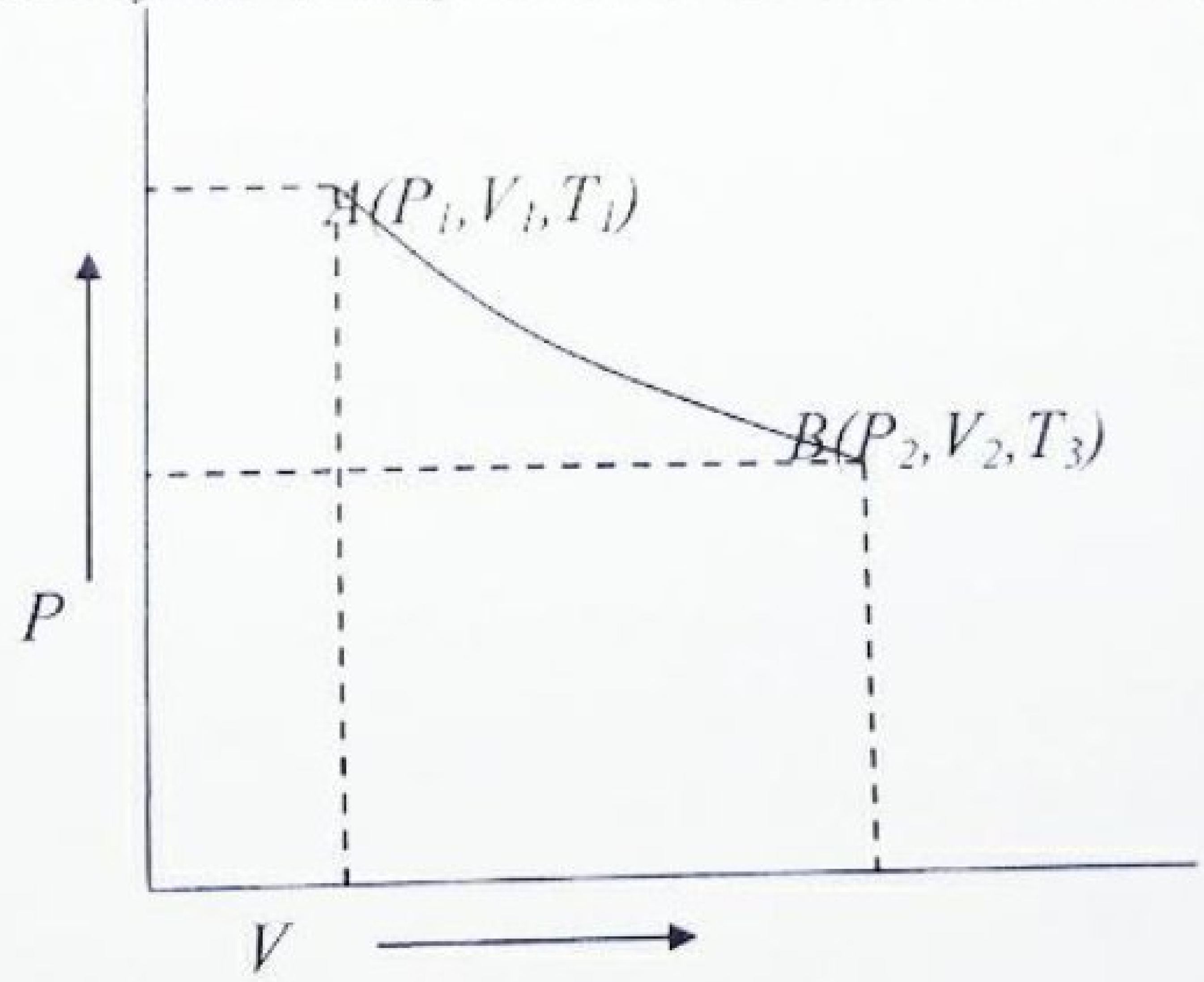
Fig.-1

Consider a working substance at a certain pressure and temperature and having a volume represented by the point A (Fig.-01).

Pressure is decreased and work is done by the working substance at the cost of its internal energy and there should be fall in temperature. But, the system is perfectly conducting to the surroundings. It absorbs heat from the surrounding and maintains a constant temperature. Thus from A to B the temperature remains constant .The curve AB is called the isothermal curve or isothermal.

Adiabatic Process:

When a system undergoes from an initial state to a final state in such a way that no heat leaves or enters the system, the process is called adiabatic process.



Thus, during an adiabatic process, the working substance is perfectly insulated from the surroundings. All along the process, there is change in temperature .A curve between pressure and volume during the adiabatic process is called an adiabatic curve or an adiabatic.

Heat Engines:

Any practical machine which converts heat into mechanical work is called a heat engine.

Heat engines in their operation absorbs heat at a higher temperature, converts part of it into mechanical work ,and rejects the remaining heat at a low temperature.

In this process, a working substance is used.In steam engines, the working substance is water vapour, and in all gas engines the working substance is a combustible mixture of gases.

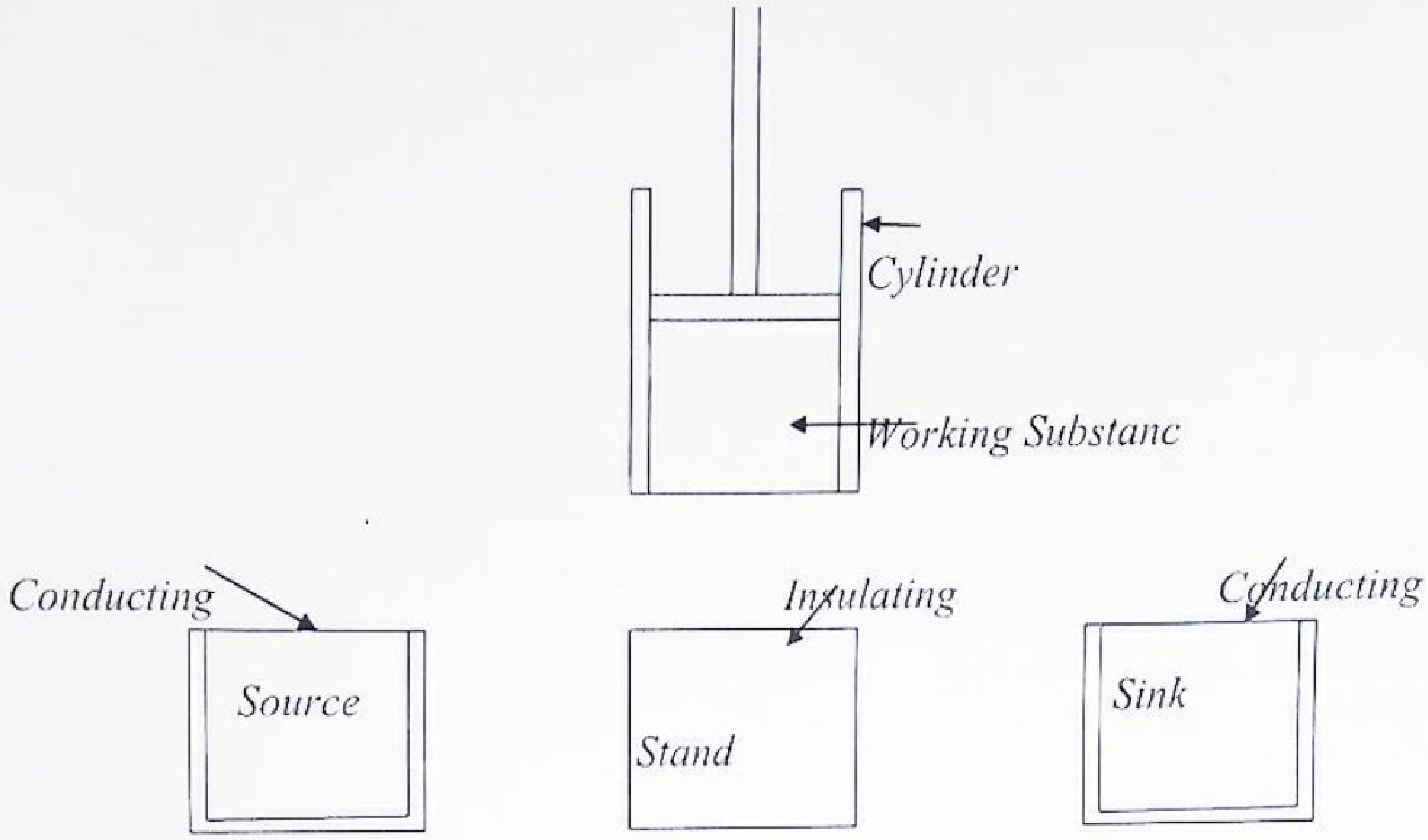
Definition of Efficiency:

The efficiency,η of a heat engine is defined as the ratio of mechanical work done by the engine in one cycle to the heat absorbed from the high temperature source. Thus, if W be the amount of work obtainable from a heat engine in one cycle at the expense of Q amount of heat, then its efficiency, η is given by

$$\eta = \frac{W}{Q},$$

Carnot's Ideal Heat Engine:

In 1824 the French engineer Sadi Carnot conceived a theoretical engine which is free from all practical imperfections. Such an engine can not be realized in practice. It has maximum efficiency and it is an ideal heat engine. Sadi Carnot's heat engine requires the following important parts (Fig.-01)



Cylinder: *A cylinder having perfectly non-conducting walls, a perfect conducting base and is provided with a perfectly non-conducting piston which moves without friction in the cylinder. The cylinder contains one mole of perfect gas as the working substance.*

Source:

A reservoir maintained at a constant temperature T_1 from which the engine can draw heat perfect conduction. It has infinite thermal capacity and any amount of heat can be drawn from it at constant temperature T_1.

Heat Insulating Stand:

A perfectly non-conducting platform acts as a stand for adiabatic process.

Sink:

A reservoir maintained at a constant temperature T_2 ($T_2 < T_1$) to which the heat engine can reject any amount of heat. The thermal capacity of sink is infinite so that its temperature remains constant at T_2, no matter how much heat is given to it.

The Carnot Cycle:

A cycle in which the working substance starting from a given condition of temperature, pressure and volume is made to undergo two successive expansion (one isothermal and another adibatic), and then brought back finally to its initial condition, is called Carnot's Cycle (Fig.-1).

Carnot's Cycle ((Heat and thermodynamics- N. Subrahmanyam, Brijlal))

In order to obtain continuous supply of work, the working substance is subjected to the following cycle of quasi-static operations known as Carnot's cycle

1. Isothermal expansion:

The cylinder is first placed on the source, so that the gas acquires the temperature T_1 of the source. It is then allowed to undergo quasi-static expansion. As the gas expands, its temperature tends to fall. Heat passes into the cylinder through the perfectly conducting base which is in contact with the source. The gas therefore, undergoes slow isothermal expansion at the constant temperature T_1.

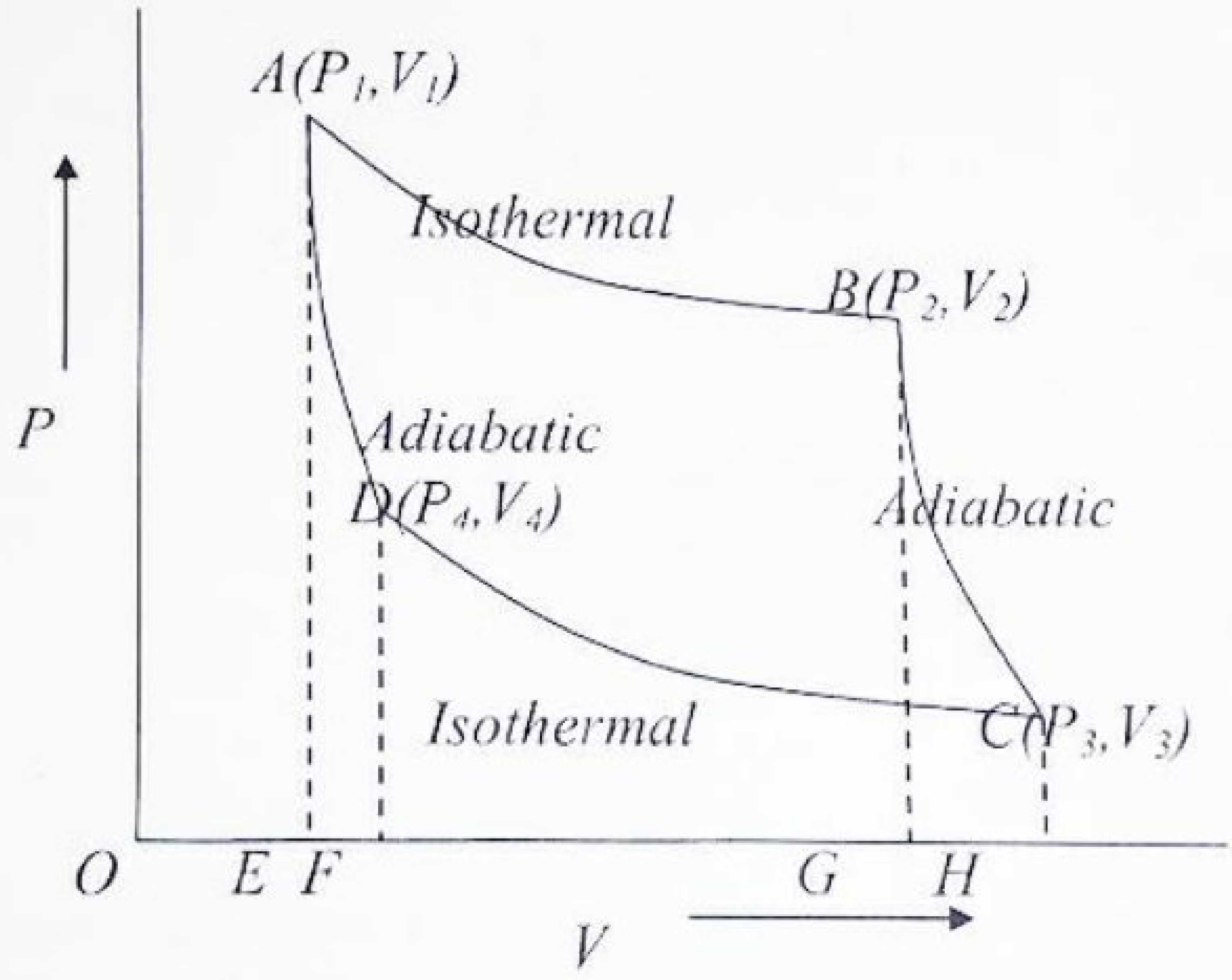


Fig.-1

Let the working substance during isothermal expansion goes from its initial state $A(P_1,V_1,T_1)$ to the state $B(P_2,V_2,T_1)$ at constant temperature T_1 along AB. In this process, the substance absorbs heat Q_1 from he source at T_1 and does work W_1 given by

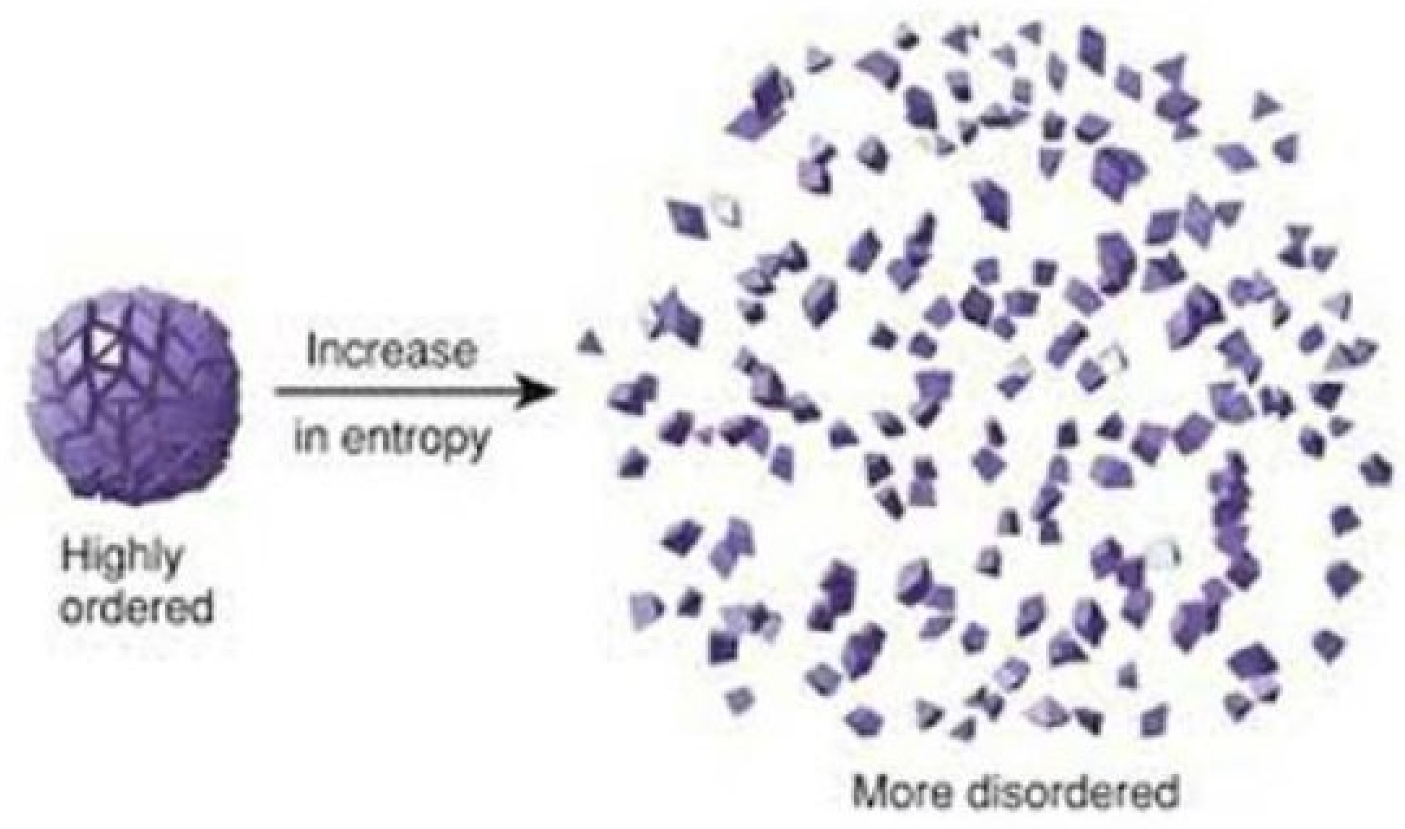
$$Q_1 = W_1 = \int_{V_1}^{V_2} PdV = RT_1 log_e \frac{V_2}{V_1} = area\ ABGEA$$

ENTROPY:

In thermodynamics, entropy (usual symbol S) is a measure of the number of specific realizations or microstates that may realize a thermodynamic system in a defined state specified by macroscopic variables. Most understand entropy as a measure of molecular disorder within a macroscopic system.

The SI units of entropy are J/K (joules/Kelvin).

The entropy of an object is a measure of the amount of energy which is unavailable to do work. Entropy is also a measure of the number of possible arrangements the atoms in a system can have. In this sense, entropy is a measure of uncertainty or randomness. The higher the entropy of an object, the more uncertain we are about the states of the atoms making up that object because there are more states to decide from. A law of physics says that it takes work to make the entropy of an object or system smaller; without work, entropy can never become smaller – you could say that everything slowly goes to disorder.



এনট্রপি(Entropy) হলো পরিসংখ্যানিক বলবিদ্যার একটি গুরুত্বপূর্ণ রাশি। কোন ভৌত ব্যবস্থায় বিদ্যমান বিশৃঙ্খলার মাত্রাকে এনট্রপির সাহায্যে প্রকাশ করা হয়। কোন প্রক্রিয়ায় (process) আগাগোড়াই যদি তাপীয় সাম্যাবস্থা সংরক্ষিত থাকে তাহলে সেখানে এনট্রপির মানও অপরিবর্তিত থাকে। তাপগতিবিদ্যার দ্বিতীয় স্বীকার্যে বলা হয়েছে যে, কোনো বিক্রিয়াতে(reaction) কখনোই মোট এনট্রপির মান হ্রাস পায় না। সহজভাবে বলতে গেলে, কোনোএক সিস্টেমের বিশৃঙ্খলাই হচ্ছে এন্ট্রপি।

work done and the heat produced. But the second law of thermodynamics gives the conditions under which heat can be converted into work.

6·26 Carnot's Reversible Engine

Heat engines are used to convert heat into mechanical work. Sadi Carnot (French) conceived a theoretical engine which is free from all the defects of practical engines. Its efficiency is maximum and it is an ideal heat engine.

For any engine, there are three essential requisites :

(1) **Source.** The source should be at a fixed high temperature T_1 from which the heat engine can draw heat. It has infinite thermal capacity and any amount of heat can be drawn from it at constant temperature T_1.

(2) **Sink.** The sink should be at a fixed lower temperature T_2 to which any amount of heat can be rejected. It also has infinite thermal capacity and its temperature remains constant at T_2.

(3) **Working Substance.** A cylinder with non-conducting sides and conducting bottom contains the perfect gas as the *working substance.*

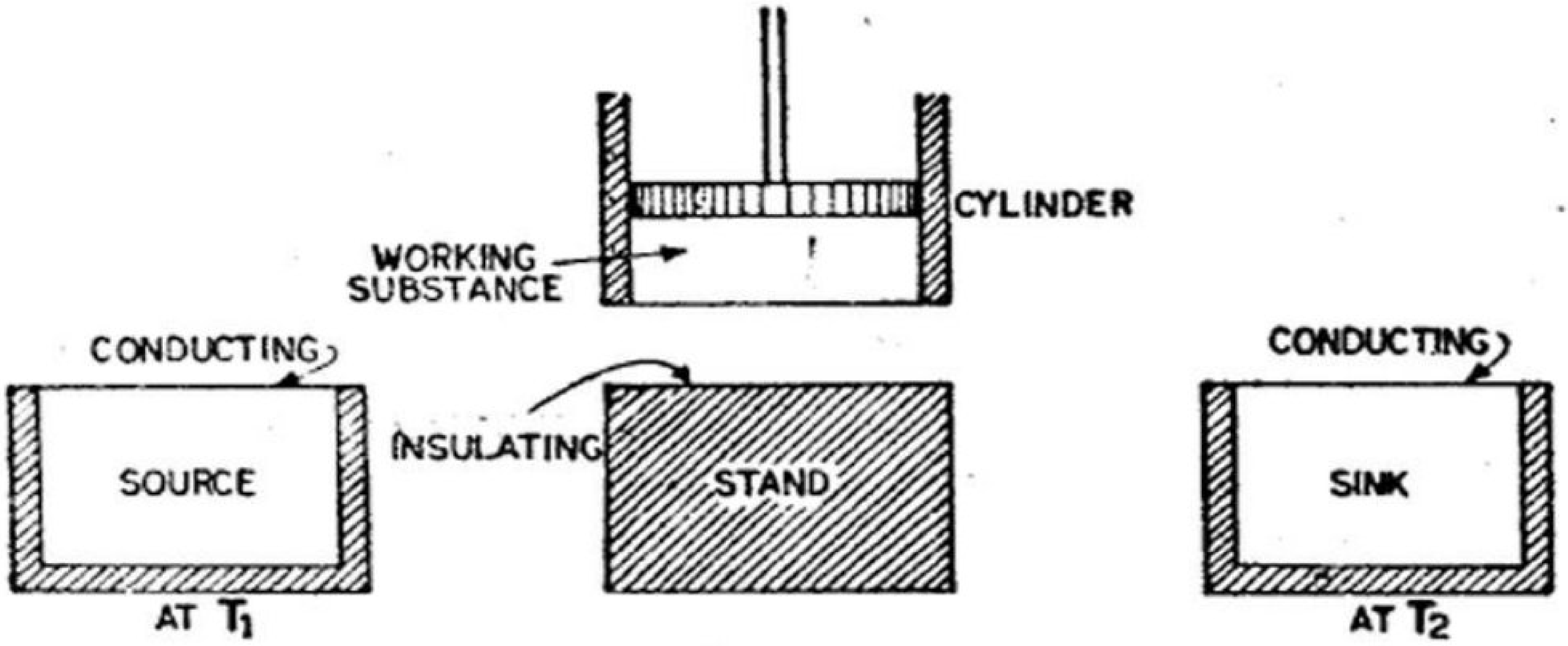


Fig. 6·12.

A perfect non-conducting and frictionless piston is fitted into the cylinder. The working substance undergoes a complete cyclic operation (Fig. 6·12).

A perfectly non-conducting stand is also provided so that the working substance can undergo adiabatic operation.

2 Carnot's Cycle

12 b and C I step 1 (1) Place the engine containing the working substance over the source at temperature T_1. The working substance is also at a temperature T_1. Its pressure is P_1 and volume is V_1 as shown by the point A in Fig. 6·13. Decrease the pressure. The volume of the working substance increases. Work is done by the working substance. As the bottom is perfectly conducting to the source at temperature T_1, it absorbs heat. The process is completely isothermal. The temperature remains constant. Let the amount of heat

absorbed by the working substance be H_1 at the temperature T_1. The point B is obtained.

Consider one gram molecule of the working substance.

Work done from A to B (isothermal process)

$$W_1 = \int_{V_1}^{V_2} P.\,dV = RT_1 \log \frac{V_2}{V_1}$$

$$= \text{area } ABGE \qquad \ldots(i)$$

(2) Place the engine on the stand having an insulated top. Decrease the pressure on the working substance. The volume

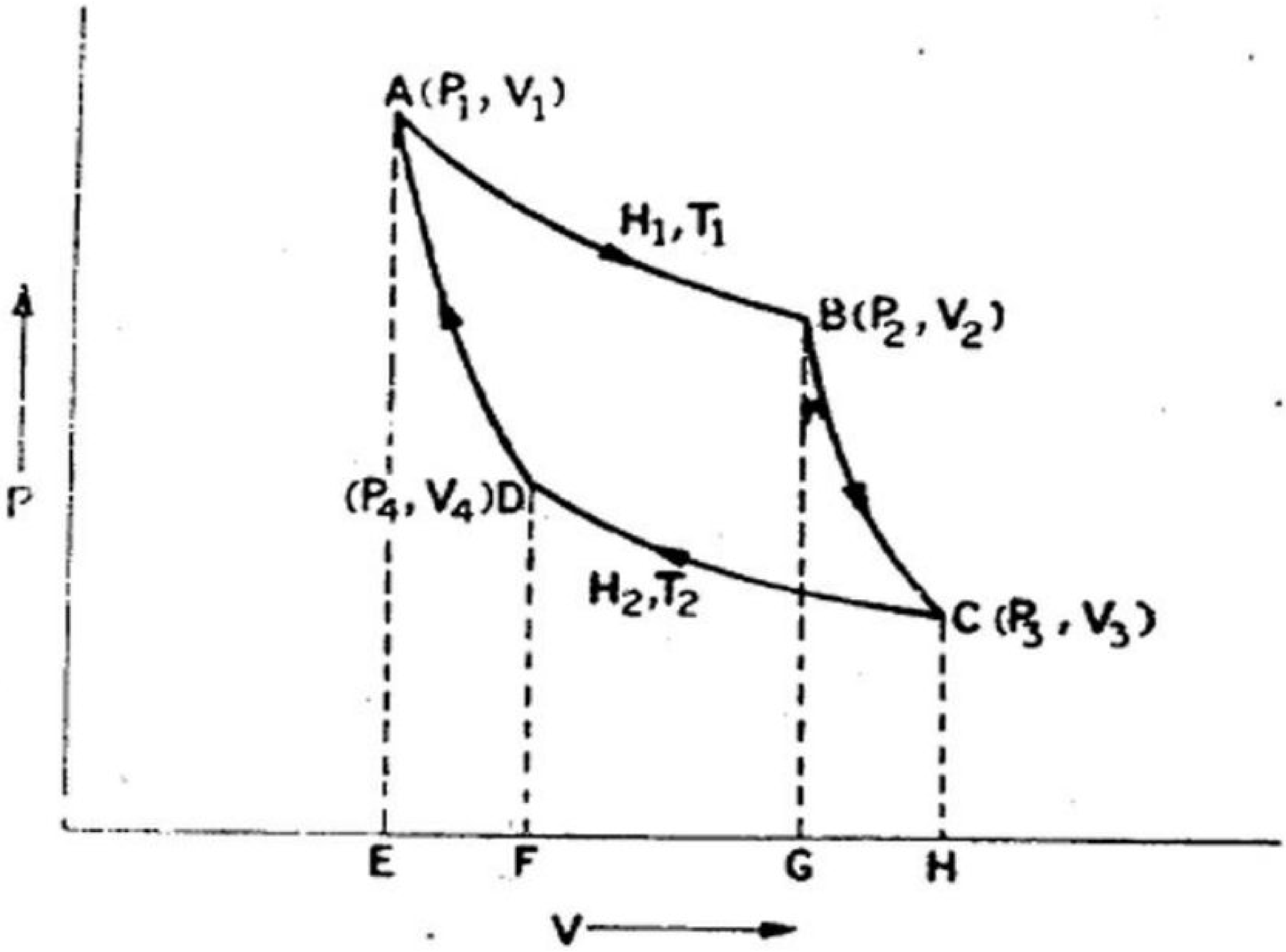


Fig. 6·13

increases. The process is completely adiabatic. Work is done by the working substance at the cost of its internal energy. The temperature falls. The working substance undergoes adiabatic change from B to C. At C the temperature is T_2 (Fig. 6·13).

Work done from B to C (adiabatic process)

$$W_2 = \int_{V_2}^{V_3} P\,.\,dV$$

$$= \int_{V_2}^{V_3} \frac{dV}{V^\gamma}$$

$$= \frac{KV_3^{1-\gamma} - KV_2^{1-\gamma}}{1-\gamma}$$

But $PV^\gamma = \text{constant} = K$

$P_2V_2 = RT_1$

$P_3V_3 = RT_2$

$P_3V_3^\gamma = P_2V_2^\gamma = K$

$$= \frac{P_3V_3 - P_2V_2}{1-\gamma}$$

$$= \frac{R[T_2 - T_1]}{1-\gamma} = \frac{R[T_1 - T_2]}{\gamma - 1}$$

$$W_2 = \text{Area } BCHG \qquad \ldots(ii)$$

(3) Place the engine on the sink at temperature T_2. Increase the pressure. The work is done on the working substance. As the base is conducting to the sink, the process is isothermal. A quantity of heat H_2 is rejected to the sink at temperature T_2. Finally the point D is reached.

Work done from C to D (isothermal process)

$$W_3 = \int_{V_3}^{V_4} PdV$$

$$= RT_2 \log \frac{V_4}{V_3}$$

$$= -RT_2 \log \frac{V_3}{V_4} \qquad \text{...(iii)}$$

$$W_3 = \text{area } CHFD$$

(The $-$ve sign indicates that work is done on the working substance.)

(4) Place the engine on the insulating stand. Increase the pressure. The volume decreases. The process is completely adiabatic. The temperature rises and finally the point A is reached.

Work done from D to A (adiabatic process).

$$W_4 = \int_{V_4}^{V_1} PdV$$

$$= -\frac{R(T_1-T_2)}{\gamma-1}$$

$$W_4 = \text{Area } DFEA \qquad \text{...(iv)}$$

[W_2 and W_4 are equal and opposite and cancel each other.] 12 b end

The net work done by the working substance in one complete cycle

$$= \text{Area } ABGE + \text{Area } BCHG - \text{Area } CHFD - \text{Area } DFEA$$

$$= \text{Area } ABCD$$

The net amount of heat absorbed by the working substance

$$= H_1 - H_2$$

$$\text{Net work} = W_1 + W_2 + W_3 + W_4$$

$$= RT_1 \log \frac{V_2}{V_1} + \frac{R(T_1-T_2)}{\gamma-1} - RT_2 \log \frac{V_3}{V_4} - \frac{R[T_1-T_2]}{\gamma-1}$$

$$W = RT_1 \log \frac{V_2}{V_1} - RT_2 \log \frac{V_3}{V_4} \qquad \text{...(v)}$$

The points A and D are on the same adiabatic

$$T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_4}\right)^{\gamma-1} \qquad \text{...(vi)}$$

The points B and C are on the same adiabatic

$$T_1V_2^{\gamma-1} = T_2V_3^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_3}\right)^{\gamma-1} \qquad ...(vii)$$

From (*vi*) and (*vii*)

$$\left(\frac{V_1}{V_4}\right)^{\gamma-1} = \left(\frac{V_2}{V_3}\right)^{\gamma-1}$$

or
$$\frac{V_1}{V_4} = \frac{V_2}{V_3}$$

or
$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

From equation (*v*)

$$W = RT_1 \log\frac{V_2}{V_1} - RT_2 \log\frac{V_2}{V_1}$$

$$W = R\left[\log\frac{V_2}{V_1}\right]\left[T_1 - T_2\right]$$

$$\therefore \quad W = H_1 - H_2$$

Efficiency $\quad \eta = \dfrac{\text{Useful output}}{\text{Input}} = \dfrac{W}{H_1}$

Heat is supplied from the source from A to B only.

$$H_1 = RT_1 \log\frac{V_2}{V_1}$$

$$\therefore \quad \eta = \frac{W}{H_1} = \frac{H_1 - H_2}{H_1}$$

$$= \frac{R[T_1 - T_2]\log\left(\frac{V_2}{V_1}\right)}{RT_1\log\left(\frac{V_2}{V_1}\right)} \quad = \frac{T_1 - T_2}{T_1}$$

$$= 1 - \frac{T_2}{T_1}$$

or
$$\eta = 1 - \frac{H_2}{H_1}$$

$$\eta = 1 - \frac{T_2}{T_1} \qquad ...(viii)$$

The Carnot's engine is perfectly reversible. It can be operated in the reverse direction also. Then it works as a refrigerator. The heat H_2 is taken from the sink and external work is done on the working substance and heat H_1 is given to the source at a higher temperature.

The isothermal process will take place only when the piston moves very slowly to give enough time for the heat transfer to take place. The adiabatic process will take place when the piston moves extremely fast to avoid heat transfer. Any practical engine cannot satisfy these conditions.

All practical engines have an efficiency less than the Carnot's engine.

6·27 Carnot's Engine and Refrigerator

Carnot's cycle is perfectly reversible. It can work as a heat engine and also as a refrigerator. When it works as a heat engine, it absorbs a quantity of heat H_1 from the source at a temperature T_1, does an amount of work W and rejects an amount of heat H_2 to the sink at temperature T_2. When it works as a refrigerator, it absorbs heat H_2 from the sink at temperature T_2. W amount of work is done on it by some external means and rejects heat H_1 to the source at a temperature T_1 (Fig. 6·14). In the second case heat flows from a body at a lower temperature to a body at a higher temperature, with the help of external work done on the working substance and it works as a refrigerator. This will not be possible if the cycle is not completely reversible.

Coefficient of Performance. The amount of heat absorbed at the lower temperature is H_2. The amount of work done by the external process (input energy) $= W$ and the amount of heat rejected $= H_1$. Here H_2 is the desired refrigerating effect.

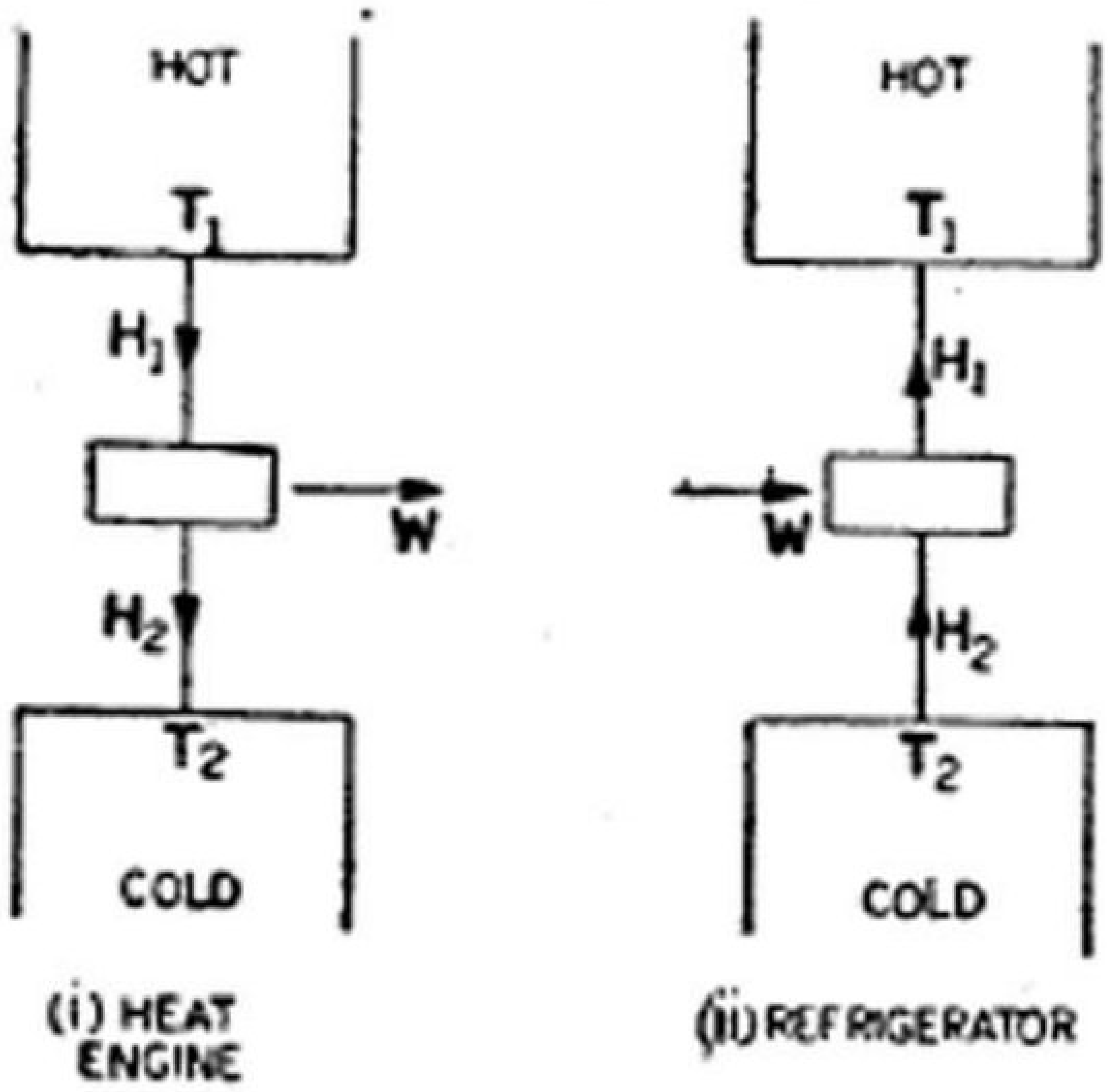


Fig. 6·14

Coefficient of performance

$$= \frac{H_2}{W} = \frac{H_2}{H_1 - H_2}$$

Suppose 200 joules of energy is absorbed at the lower temperature and 100 joules of work is done with external help. Then $200 + 100 = 300$ joules are rejected at the higher temperature.

The coefficient of preformance

$$= \frac{H_2}{W}$$

$$= \frac{H_2}{H_1 - H_2}$$

$$= \frac{200}{300 - 200} = 2$$

Therefore the coefficient of performance of a refrigerator = 2.

In the case of a heat engine, the efficiency cannot be more than 100% but in the case of a refrigerator, the coefficient of performance can be much higher than 100%.

Example 6·8. *Find the efficiency of the Carnot's engine working between the steam point and the ice point.*

$$T_1 = 273 + 100 = 373 \text{ K}$$

$$T_2 = 273 + 0 = 273 \text{ K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{273}{373} = \frac{100}{373}$$

$$\% \text{ efficiency} = \frac{100}{373} \times 100$$

$$= \mathbf{26{\cdot}81\%}$$

Example 6·9. *Find the efficiency of a Carnot's engine working between 127°C and 27°C.*

$$T_1 = 273 + 127 = 400 \text{ K}$$

$$T_2 = 273 + 27 = 300 \text{ K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{400} = 0{\cdot}25$$

$$\% \text{ efficiency} = \mathbf{25\%}$$

Example 6·10. *A Carnot's engine whose temperature of the source is 400 K takes 200 calories of heat at this temperature and rejects 150 calories of heat to the sink. What is the temperature of the sink ? Also calculate the efficiency of the engine.*

$H_1 = 200$ cal ; $H_2 = 150$ cal

$T_1 = 400$ K ; $T_2 = ?$

$$\frac{H_1}{T_1} = \frac{H_2}{T_2}$$

$$T_2 = \frac{H_2}{H_1} \times T_1$$

$$T_2 = \frac{150}{200} \times 400 = \mathbf{300 \text{ K}}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{400} = 0{\cdot}25$$

% efficiency = **25%**

Example 6·11. *A Carnot's engine is operated between two reservoirs at temperatures of 450* K *and 350* K. *If the engine receives 1000 calories of heat from the source in each cycle, calculate the amount of heat rejected to the sink in each cycle. Calculate the efficiency of the engine and the work done by the engine in each cycle.* (*1* calorie = *4·2* joules).

$$T_1 = 450 \text{ K}; \qquad T_2 = 350 \text{ K}$$

$$H_1 = 1000 \text{ cal}; \qquad H_2 = ?$$

$$\frac{H_2}{H_1} = \frac{T_2}{T_1}$$

$$H_2 = H_1 \times \frac{T_2}{T_1}$$

$$= \frac{1000 \times 350}{450} = \mathbf{777{\cdot}77 \text{ cals}}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{350}{450} = \frac{100}{450}$$

$$= 0{\cdot}2222$$

% efficiency = 22·22%

Work done in each cycle

$$= H_1 - H_2$$

$$= 1000 - 777{\cdot}77$$

$$= 222{\cdot}23 \text{ cal}$$

$$= 222{\cdot}23 \times 4{\cdot}2 \text{ joules}$$

$$= \mathbf{933{\cdot}33 \text{ joules}}$$

Example 6·12. *A Carnot's engine working as a refrigerator between 260* K *and 300* K *receives 500* calories *of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected to the reservoir at the higher temperature. Calculate also the amount of work done in each cycle to operate the refrigerator.*

[*Delhi* (*Hons.*) *1974*]

$$H_1 = ? \qquad H_2 = 500 \text{ cal}$$

$$T_1 = 300 \text{ K} \qquad T_2 = 260 \text{ K}$$

$$\frac{H_1}{H_2} = \frac{T_1}{T_2};$$

$$H_1 = H_2 \cdot \frac{T_1}{T_2}$$

$$H_1 = \frac{500 \times 300}{260} = 576{\cdot}92 \text{ cal}$$

$$W = H_1 - H_2 = 76{\cdot}92 \text{ cal}$$

$$= 76{\cdot}92 \times 4{\cdot}2 \text{ joules}$$

$$= \mathbf{323{\cdot}08 \text{ joules}}$$

Example 6·13. *A Carnot's refrigerator takes heat from water at 0°C and discards it to a room at 27°C. 1 kg of water at 0°C is to be changed into ice at 0°C. How many calories of heat are discarded to the room? What is the work done by the refrigerator in this process? What is the coefficient of performance of the machine?*

[*Delhi 1974*]

$$H_1 = ?$$

$$H_2 = 1000 \times 80 = 80{,}000 \text{ cal}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = 273 \text{ K}$$

(1)
$$\frac{H_1}{H_2} = \frac{T_1}{T_2}$$

$$H_1 = \frac{H_2 T_1}{T_2}$$

$$= \frac{80{,}000 \times 300}{273}$$

$$H_1 = \mathbf{87{,}900 \text{ Cal}}$$

(2) Work done by the refrigerator

$$= W = J(H_1 - H_2)$$

$$W = 4{\cdot}2\,(87{,}900 - 80{,}000)$$

$$W = 4{\cdot}2 \times 7900$$

or
$$W = \mathbf{3{\cdot}183 \times 10^4 \text{ joules}}$$

(3) Coefficient of performance,

$$= \frac{H_2}{H_1 - H_2}$$

$$= \frac{80{,}000}{87{,}900 - 80{,}000}$$

$$= \frac{80{,}000}{7900}$$

$$= \mathbf{10{\cdot}13}$$

Example 6·14. *A carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased ?* (Delhi 1971)

In the first case

$$\eta = 50\% = 0{\cdot}5, \quad T_2 = 273+7 = 280 \text{ K.}$$

$$T_1 = ?$$

$$\eta = 1 - \frac{T_2}{T_1}$$

or
$$0{\cdot}5 = 1 - \frac{280}{T_1}$$

or
$$T_1 = 560\ K$$

In the second case

$$\eta' = 70\% = 0{\cdot}7,$$

$$T_2 = 280\ K,$$

$$T_1' = ?$$

$$\eta' = 1 - \frac{T_2}{T_1'}$$

$$0{\cdot}7 = 1 - \frac{280}{T_1'}$$

or
$$T_1' = 840\ K$$

Increase in temperature = 840−560 = **280 K**

6·28 Carnot's Theorem

The efficiency of a reversible engine does not depend on the nature of the working substance. It merely depends upon the temperature limits between which the engine works.

Carnot's Theorem 12 Short

"All the reversible engines working between the same temperature limits have the same efficiency. No engine can be more efficient than a Carnot's reversible engine working between the same two temperatures."

Consider two reversible engines A and B, working between the temperature limits T_1 and T_2 (Fig. 6·15). A and B are coupled. Suppose A is more efficient than B. The engine A works as a heat engine and B as a refrigerator. The engine A absorbs an amount of heat H_1 from the source at a temperature T_1. It does external work W and transfers it to B. The heat rejected to the sink is H_2 at a temperature T_2. The engine B absorbs heat H_2' from the sink at temperature T_2 and W amount of work is done on the working substance. The heat given to the source at temperature T_1 is H_1'.

Suppose the engine A is more efficient than B.

Efficiency of the engine A

$$= \eta = \frac{H_1 - H_2}{H_1} = \frac{W}{H_1}$$

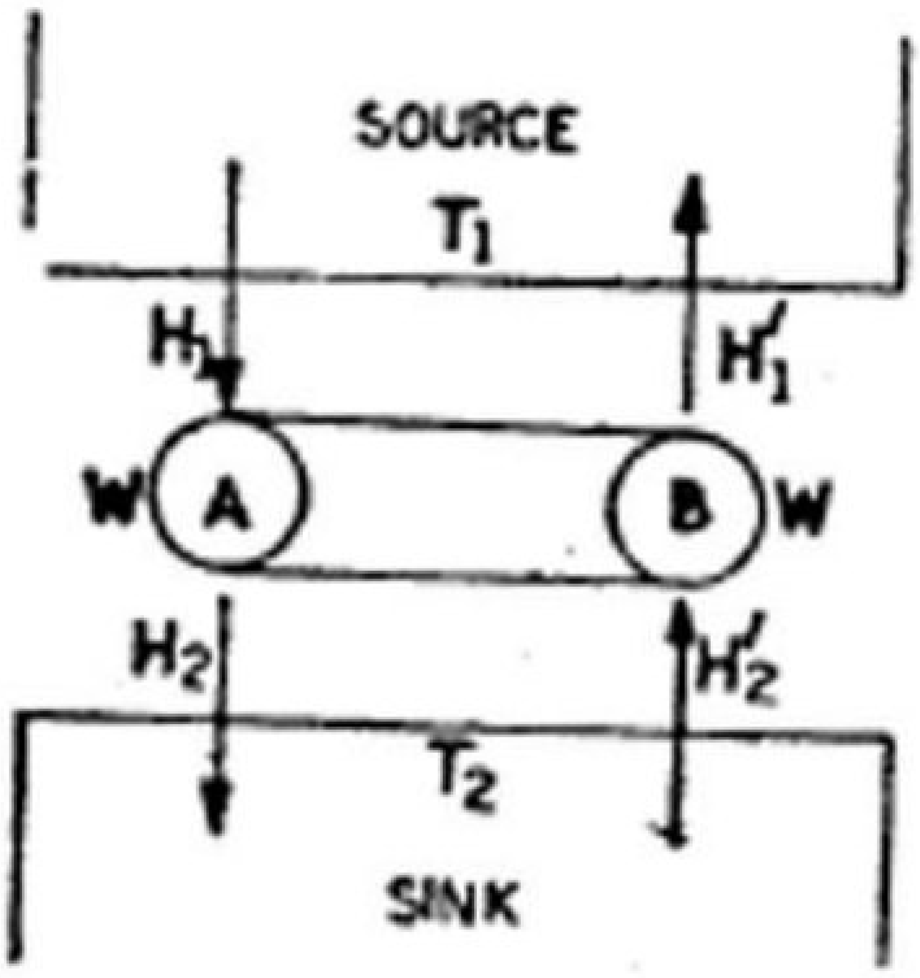


Fig. 6·15.

Efficiency of the engine B

$$= \eta' = \frac{H_1' - H_2'}{H_1'} = \frac{W}{H_1'}$$

Since $\qquad \eta > \eta'$; $H_1' > H_1$

Also, $\qquad W = H_1 - H_2 = H_1' - H_2'$

$\therefore \qquad H_2' > H_2$

Thus, for the two engines A and B working as a coupled system, $(H_2' - H_2)$ is the quantity of heat taken from the sink at a temperature T_2 and $(H_1' - H_1)$ is the quantity of heat given to the source at a temperature T_1. Both $(H_2' - H_2)$ and $(H_1' - H_1)$ are positive quantities. It means heat flows from the sink at a temperature T_2 (lower temperature) to the source at a temperature T_1 (higher temperature) *i.e.*, heat flows from a body at a lower temperature to a body at a higher temperature. But, no external work has been done on the system. This is contrary to the second law of thermodynamics. Thus, η cannot be greater than η'. The two engines (reversible) working between the same two temperature limits have the same efficiency. Moreover, in the case of a Carnot's engine, there is no loss of heat due to friction, conduction or radiation (irreversible processes). Thus, the Carnot's engine has the maximum efficiency. Whatever may be the nature of the working substance, the efficiency depends only upon the two temperature limits.

In a practical engine there is always loss of energy due to friction, conduction, radiation etc. and hence its efficiency is always lower than that of a Carnot's engine.

finish 2nd.

Example 6·15. *An inventor claims to have developed an engine working between 600 K and 300 K capable of having an efficiency of 52%. Comment on his claim.*

The efficiency of a Carnot's engine working between 600 K and 300 K,

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{600} = 0{\cdot}5$$

$$= 50\%$$

The efficiency claimed $= \eta = 52\%$

It means that the efficiency of the engine is more than the efficiency of a Carnot's engine working between the same two temperature limits. But, no engine can have an efficiency more than a Carnot's engine, so his claim is invalid.

6·29 Thermodynamic (or Work or Absolute) Scale of Temperature

The efficiency of a Carnot's engine does not depend upon the property of the working substance but it depends upon the temperature limits between which the engine works. The scale of temperature based on the working of the Carnot's engine is a standard scale and it does not depend upon the particular property of any substance, as in the case of other thermometric scales. Kelvin worked out the theory of the absolute scale called Kelvin's or thermodynamical scale and also showed that it agrees with the perfect gas scale.

Suppose an engine works between the temperatures θ_1 and θ_2. Here H_1 is the heat absorbed at θ_1 and H_2 is the heat rejected at θ_2

$$\eta = f(\theta_1, \theta_2)$$

$$\eta = \frac{H_1 - H_2}{H_1} = f(\theta_1, \theta_2)$$

$$\therefore \quad 1 - \frac{H_1}{H_2} = f(\theta_1, \theta_2)$$

or

$$\frac{H_1}{H_2} = \frac{1}{1 - f(\theta_1, \theta_2)} = F(\theta_1, \theta_2) \quad \text{...}(i)$$

Similarly, if the engine works between the temperature θ_2 and θ_3 and H_2 is the heat absorbed and H_3 is the heat rejected, then

$$\frac{H_2}{H_3} = F(\theta_2, \theta_3) \quad \text{...}(ii)$$

If the engine works between the temperatures θ_1 and θ_3 and H_1 is the heat absorbed and H_3 is the heat rejected, then

$$\frac{H_1}{H_3} = F(\theta_1, \theta_3) \quad \text{...(iii)}$$

From (*i*), (*ii*) and (*iii*),

$$\frac{H_1}{H_3} = \frac{H_1}{H_2} \times \frac{H_2}{H_3}$$

$$\therefore \quad F(\theta_1, \theta_3) = F(\theta_1, \theta_2) \times F(\theta_2, \theta_3)$$

$$F(\theta_1, \theta_2) = \frac{F(\theta_1, \theta_3)}{F(\theta_2, \theta_3)}$$

This temperature θ_3 is arbitrarily chosen.

This relation can be satisfied only if

$$F(\theta_1, \theta_2) = \frac{\phi(\theta_1)}{\phi(\theta_2)}$$

$$F(\theta_2, \theta_3) = \frac{\phi(\theta_2)}{\phi(\theta_3)}$$

and

$$F(\theta_1, \theta_3) = \frac{\phi(\theta_1)}{\phi(\theta_3)}$$

where ϕ is another function.

$$\therefore \quad \frac{H_1}{H_2} = F(\theta_1, \theta_2) = \frac{\phi(\theta_1)}{\phi(\theta_2)}$$

The expression on the right hand side represents the ratio of the two Kelvin temperatures and this can be denoted as θ_1/θ_2.

$$\therefore \quad \frac{H_1}{H_2} = \frac{\theta_1}{\theta_2} \quad \text{...(iv)}$$

The relation (*iv*) is used to represent a new scale and this does not depend upon the property of the working substance.

$$\text{Efficiency} \quad \eta = \frac{H_1 - H_2}{H_1} = \frac{\theta_1 - \theta_2}{\theta_1} \quad \text{...(v)}$$

$$\eta = 1 - \frac{\theta_2}{\theta_1} \quad \text{...(vi)}$$

6·30 Absolute Zero on Work Scale

Consider an engine working between the steam point and the ice point. The isothermal *AB* is at the steam point and the isothermal *OD* is at the ice point (Fig. 6·16).

$$\eta = 1 - \frac{\theta_{ice}}{\theta_{steam}}$$

$$\eta = \frac{\theta_{steam} - \theta_{ice}}{\theta_{steam}}$$